

The Chevalley commutator with the height-graded ribbon operator is $(1 + qt)$ times a unit monomial: a closed form

Clio

31 August 2026

Abstract

Let $\mathcal{F}$ be the level-1 q -deformed Fock space of Hayashi–Misra–Miwa, with its $U_q(\widehat{\mathfrak{sl}}_e)$ -action, and let $R_e(t) = \sum_{h=0}^{e-1} t^h N_e^{(h)}$ be the generating operator of e -ribbon addition graded by ribbon height (spin). We compute, in closed form, every matrix entry of $[x, R_e(t)]$ for x a Chevalley generator e_i or f_i . The answer is

$$([x, R_e(t)])_{\nu\lambda} = \varepsilon q^k t^b (1 + qt), \quad \varepsilon \in \{0, \pm 1\}, \quad k \in \mathbb{Z}, \quad 0 \leq b \leq e - 2,$$

with ε, k, b given explicitly by three bead statistics of the abacus of λ . This proves question Q59 (opened 2026-08-30) in the affirmative and in a form strictly stronger than conjectured: it determines the vanishing locus, the sign, the q -power and the t -degree, rather than merely constraining their shape. Four corollaries follow, including a new proof that $B_{-1}^{[e]} = R_e(-q^{-1})$ commutes with $U_q(\widehat{\mathfrak{sl}}_e)$, and the promotion of the previously *computed* statements (P1), (P2) and “every entry of $[x, P_e]$ is $\pm q^m (q - q^{-1})$ ” to *proved*. All statements are verified against an independent matrix computation: 7328 nonzero entries for $2 \leq e \leq 9$, $|\lambda| \leq 14$, with no discrepancy.

1 Setting and conventions

Throughout $e \geq 2$ is fixed, and q is an indeterminate; $\mathbb{Z}[q^{\pm 1}]$ acts on everything. Partitions are weakly decreasing finite sequences of positive integers, identified with their Young diagrams in English (matrix) orientation; the cell in row r and column c has *content* $c - r$ and *residue* $(c - r) \bmod e$.

Level-1 q -Fock space is $\mathcal{F} = \bigoplus_{\lambda} \mathbb{Z}[q^{\pm 1}]|\lambda\rangle$, the sum over all partitions. We use the Leclerc–Thibon normalisation of the $U_q(\widehat{\mathfrak{sl}}_e)$ -action. For γ an addable or removable i -node of π , set

$$N_i^{\downarrow}(\pi, \gamma) = \#\{\text{addable } i\text{-nodes of } \pi \text{ strictly below } \gamma\} - \#\{\text{removable } i\text{-nodes of } \pi \text{ strictly below } \gamma\},$$

and $N_i^{\uparrow}(\pi, \gamma)$ for the same count taken strictly above. Then

$$f_i|\lambda\rangle = \sum_{\gamma} q^{N_i^{\uparrow}(\lambda, \gamma)} |\lambda + \gamma\rangle, \quad e_i|\lambda\rangle = \sum_{\gamma} q^{-N_i^{\downarrow}(\lambda - \gamma, \gamma)} |\lambda - \gamma\rangle, \quad (1)$$

the sums being over addable (resp. removable) i -nodes γ of λ . (This is the convention pinned down computationally in `probes/2026-08-30-Q55-transpose/chevalley.py`: it is the unique one of the four sign/side normalisations for which $[e_i, B_{-1}^{[e]}] = 0$. Note the asymmetry: the weight of the e_i step is read off the *target* partition.)

Definition 1. For $0 \leq h \leq e-1$ let $N_e^{(h)}$ be the operator $N_e^{(h)}|\lambda\rangle = \sum_{\mu} |\mu\rangle$, the sum over μ obtained from λ by adding an e -ribbon spanning exactly $h+1$ rows (i.e. of *height*, or *spin*, h). Put

$$R_e(t) = \sum_{h=0}^{e-1} t^h N_e^{(h)}.$$

Specialising t recovers the operators of the programme: $P_e = R_e(-q)$, $B_{-1}^{[e]} = R_e(-q^{-1})$, and $C_e^{(1)} = \sum_h (-1)^h [h]_q N_e^{(h)}$.

The object of study is

$$\Phi_{\nu\lambda}^{x,i}(t) := ([x_i, R_e(t)])_{\nu\lambda} \in \mathbb{Z}[q^{\pm 1}][t], \quad \deg_t \leq e-1,$$

for $x \in \{e, f\}$. Q59 asked whether Φ is always 0 or $\varepsilon q^k t^b (1+qt)$ with $\varepsilon = \pm 1$ and $0 \leq b \leq e-2$.

2 The abacus, and why it is the right language

Definition 2. The *Maya set* of a partition λ is $M(\lambda) = \{\lambda_j - j : j \geq 1\} \subset \mathbb{Z}$, where $\lambda_j = 0$ for $j > \ell(\lambda)$. It is cofinite in $\mathbb{Z}_{<0}$ and coinfinite in $\mathbb{Z}_{\geq 0}$. We write $m = m_\lambda = \mathbf{1}_{M(\lambda)}$ and call the elements of M *beads*, the elements of its complement *holes*.

The three dictionary entries we need are classical.

Lemma 3 (Dictionary). *Let $M = M(\lambda)$.*

- (i) λ has a removable node of content $p \iff p \in M$ and $p-1 \notin M$; removing it replaces M by $M \setminus \{p\} \cup \{p-1\}$.
- (ii) λ has an addable node of content $p \iff p-1 \in M$ and $p \notin M$; adding it replaces M by $M \setminus \{p-1\} \cup \{p\}$.
- (iii) The e -ribbons addable to λ are in bijection with the $b \in M$ with $b+e \notin M$; the corresponding μ has $M(\mu) = M \setminus \{b\} \cup \{b+e\}$, the ribbon's cells have contents $b+1, b+2, \dots, b+e$, and its height is $h = \#(M \cap (b, b+e))$.

In particular, for each content p the partition λ has at most one node (addable or removable) of content p , and

$$\chi_\lambda(p) := [\text{addable at } p] - [\text{removable at } p] = m(p-1) - m(p) \quad \text{for every } p \in \mathbb{Z}. \quad (2)$$

Proof. (i)–(iii) are the standard abacus dictionary; (iii) is the implementation used throughout this programme, cross-checked against a direct diagram enumeration (`ribbons.py`: 239 agreements at $e \leq 6$, and 3556 partitions at $e \leq 8$, $|\lambda| \leq 14$, zero failures). For (2): if $m(p-1) = m(p)$ there is neither an addable nor a removable node of content p by (i),(ii), and the right-hand side is 0; the two remaining cases are exactly (i) and (ii). $\square$

The next lemma is the engine of the whole paper: it turns the Chevalley exponents, which are defined by a *geometric* condition (“strictly below”), into sums of bead indicators along an arithmetic progression of step e .

Lemma 4 (Telescope). *Let γ be an addable or removable node of π , of content c , so that $i \equiv c \pmod{e}$. Then, with $m = m_\pi$,*

$$N_i^\downarrow(\pi, \gamma) = F_\pi(c) := \sum_{k \geq 1} (m(c - ke - 1) - m(c - ke)), \quad N_i^\uparrow(\pi, \gamma) = G_\pi(c) := \sum_{k \geq 1} (m(c + ke - 1) - m(c + ke)).$$

Both sums have finitely many nonzero terms.

Proof. First, among the addable and removable nodes of π , “strictly below γ ” coincides with “of content $< c$ ” once we restrict to nodes of residue i . Indeed, the nodes lying in row r are (r, π_r) (removable, content $\pi_r - r$, present iff $\pi_r > \pi_{r+1}$) and $(r, \pi_r + 1)$ (addable, content $\pi_r + 1 - r$, present iff $r = 1$ or $\pi_{r-1} > \pi_r$). Comparing rows r and $r + 1$: every node in row $r + 1$ has content at most $\pi_{r+1} - r$, while every node in row r has content at least $\pi_r - r$; and $\pi_{r+1} - r \leq \pi_r - r$ with equality only if $\pi_{r+1} = \pi_r$, in which case row r has no removable node and its unique node has content $\pi_r + 1 - r > \pi_{r+1} - r$. So contents of nodes strictly decrease as the row index increases. The only nodes not separated by this argument are the two nodes of a single row, whose contents differ by 1; since $e \geq 2$ they have different residues, so they are never compared when we fix the residue i .

Hence $N_i^\downarrow(\pi, \gamma) = \sum_{p < c, p \equiv i \pmod{e}} \chi_\pi(p)$, and substituting (2) with $p = c - ke$, $k \geq 1$, gives the stated formula. The $N_i^\uparrow$ case is identical with $p = c + ke$. Finiteness holds because $m \equiv 1$ sufficiently far below and $m \equiv 0$ sufficiently far above. $\square$

Remark 5. Lemma 4 was tested against a direct implementation of $N_i^\downarrow, N_i^\uparrow$ for all addable and removable nodes of all λ with $|\lambda| \leq 10$ and $2 \leq e \leq 6$: 7070 checks, 0 failures (`maya.py`).

3 Classification of the paths

A matrix entry of $x_i R_e(t)$ or $R_e(t) x_i$ is a sum over length-two paths in which one step is an e -ribbon addition (bead move $b \mapsto b + e$) and the other is a single-node move (bead move $c \mapsto c - 1$ for e_i , or $d - 1 \mapsto d$ for f_i). The classification below is nothing more than the observation that *two bead moves can compose to a net displacement of one bead in exactly two ways*, and otherwise the two moves are independent.

We treat $x = e_i$ in detail; the case $x = f_i$ is verbatim parallel and recorded in §6.

Fix λ , write $M = M(\lambda)$, $m = \mathbf{1}_M$, and fix ν with $|\nu| = |\lambda| + e - 1$. A contributing path is a pair (b, c) where the ribbon moves the bead at b to $b + e$ and the node removed has content $c \equiv i \pmod{e}$; the composite effect on Maya sets is

$$M \mapsto M \setminus \{b, c\} \cup \{b + e, c - 1\}, \quad (3)$$

read in either order.

Lemma 6 (Two regimes). *Exactly one of the following holds for (λ, ν) :*

- (I) $\nu \not\supseteq \lambda$. Then there is at most one pair (b, c) producing ν in (3), and it satisfies $\{b, b + e\} \cap \{c - 1, c\} = \emptyset$.
- (II) $\nu \supseteq \lambda$ and ν/λ is an $(e - 1)$ -ribbon, given by the bead move $a \mapsto a + e - 1$ with $a \in M$, $a + e - 1 \notin M$. Then exactly four pairs (b, c) are formally possible, namely

$$H : (b, c) = (a, a + e), \quad T : (b, c) = (a - 1, a),$$

each read in the two possible orders, giving the four path types H, T (ribbon first) and H', T' (node first).

If ν is of neither kind, $\Phi_{\nu\lambda}^{e,i} = 0$ identically.

Proof. Write (3) as: remove $\{b, c\}$, adjoin $\{b + e, c - 1\}$. Since M and its image are both Maya sets of the same charge, the multiset identity forces the removed and adjoined lists to have the same length after cancellation.

No cancellation, i.e. $b \neq c$, $b + e \neq c - 1$, $b + e \neq c$, $b \neq c - 1$: then ν is obtained from λ by moving two distinct beads, one forwards by e and one backwards by 1; in particular $c \in M \setminus M(\nu)$, so $\nu \not\supseteq \lambda$ (regime (I)). Uniqueness of the pair: suppose $M \setminus M(\nu) = \{p_1, p_2\}$ and $M(\nu) \setminus M = \{q_1, q_2\}$ admit two matchings, $\{p_1 + e, p_2 - 1\} = \{q_1, q_2\}$ and $\{p_2 + e, p_1 - 1\} = \{q_1, q_2\}$. Equating the two descriptions gives either $p_1 + e = p_2 + e$, i.e. $p_1 = p_2$, or $p_1 + e = p_1 - 1$, i.e. $e = -1$; both are absurd.

Cancellation. $b = c$ is impossible: reading ribbon-first, c must be a bead of $M' = M \setminus \{b\} \cup \{b + e\}$, and $b \notin M'$; reading node-first, b must be a bead of $M'' = M \setminus \{c\} \cup \{c - 1\}$, and $c \notin M''$. Similarly $b + e = c - 1$ is impossible, since ribbon-first requires $c - 1 \notin M' \ni b + e$ and node-first requires $b + e \notin M'' \ni c - 1$. There remain

$$b + e = c \quad (\text{type H}), \quad c - 1 = b \quad (\text{type T}),$$

and these cannot hold simultaneously (they would give $e = -1$). In type H the net effect (3) is $M \mapsto M \setminus \{b\} \cup \{b + e - 1\}$, i.e. $a = b$ and $c = a + e$; in type T it is $M \mapsto M \setminus \{c\} \cup \{c + e - 1\}$, i.e. $a = c$ and $b = a - 1$. Either way ν arises from λ by a single bead move $a \mapsto a + e - 1$, that is, by adding an $(e - 1)$ -ribbon; so $\nu \supseteq \lambda$ (regime (II)), and a is determined by (λ, ν) .

Finally, in both H and T the removed node has content $\equiv a \pmod{e}$, so $\Phi_{\nu\lambda}^{e,i} = 0$ unless $i \equiv a$. $\square$

4 Regime (I): exact cancellation

Proposition 7. *If $\nu \not\supseteq \lambda$ then $\Phi_{\nu\lambda}^{e,i}(t) = 0$.*

Proof. By Lemma 6(I) there is a single candidate pair (b, c) , with $\{b, b + e\} \cap \{c - 1, c\} = \emptyset$; if there is none, both products vanish at (ν, λ) and we are done. Write $M' = M \setminus \{b\} \cup \{b + e\}$ (Maya set of $\mu = \lambda + \rho$) and $M'' = M \setminus \{c\} \cup \{c - 1\}$ (Maya set of $\kappa = \lambda - \gamma$).

Legality. The ribbon-first path is legal iff $b \in M$, $b + e \notin M$ and $(c \in M', c - 1 \notin M')$. Because M' agrees with M off $\{b, b + e\}$ and $c, c - 1 \notin \{b, b + e\}$, the bracketed condition is equivalent to $c \in M$, $c - 1 \notin M$. The same computation on the node-first path gives the same four conditions. So the two paths exist simultaneously; they contribute the same power of t , namely t^h with $h = \#(M \cap (b, b + e))$ in the first order and $h'' = \#(M'' \cap (b, b + e))$ in the second — and M'' agrees with M off $\{c - 1, c\} \subseteq \mathbb{Z} \setminus \{b, b + e\}$; moreover $c - 1, c$ lie in $(b, b + e)$ or not simultaneously (they differ by 1 and neither equals b or $b + e$), and M'' has one bead in $\{c - 1, c\}$ exactly as M does. Hence $h'' = h$.

Weights. The two contributions are $q^{-N^\downarrow(\nu, \gamma)}$ (ribbon first, the e_i -step being read on its target ν) and $q^{-N^\downarrow(\kappa, \gamma)}$ (node first), with opposite signs in the commutator. By Lemma 4 their exponents are $F_\nu(c)$ and $F_\kappa(c)$; and $M(\nu)$ is obtained from $M'' = M(\kappa)$ by the bead move $b \mapsto b + e$. So it suffices to show that this bead move does not change $F(c)$.

$F(c)$ involves m only at positions $c - ke - 1$ ($k \geq 1$), all $\equiv c - 1$ and all $< c - 1$, and $c - ke$ ($k \geq 1$), all $\equiv c$ and all $< c$. If $b \not\equiv c, c - 1 \pmod{e}$ neither b nor $b + e$ occurs and F is unchanged. If $b \equiv c$: since $b \neq c$ and $b + e \neq c$, either $b < c$ — and then $b \leq c - e$, so $b + e \leq c$, hence $b + e < c$ and both b and $b + e$ occur in the sum, contributing $+1$ and -1 — or $b > c$, and then $b + e > c$

too and neither occurs. Either way $\Delta F = 0$. If $b \equiv c - 1$ the same dichotomy at $c - 1$ gives $\Delta F = -1 + 1 = 0$ or 0. Hence the two contributions are equal and cancel in the commutator. $\square$

5 Regime (II): the closed form

Fix λ , $M = M(\lambda)$, and $a \in M$ with $a + e - 1 \notin M$; let ν be given by $M(\nu) = M \setminus \{a\} \cup \{a + e - 1\}$, and take $i \equiv a \pmod{e}$. Set

$$\alpha := m(a-1) \in \{0, 1\}, \quad \beta := m(a+e) \in \{0, 1\}, \quad s := \#(M \cap [a+1, a+e-2]), \quad \phi := F_\lambda(a). \quad (4)$$

Note $0 \leq s \leq e - 2$.

Lemma 8 (The four types). *With the notation of Lemma 6(II):*

type	order	legal iff	height	weight
H	ribbon, then e_i	$\beta = 0$	s	$+q^{-\phi-\alpha}$
T	ribbon, then e_i	$\alpha = 1$	$s + 1$	$+q^{-\phi}$
H'	e_i , then ribbon	$\beta = 1$	$s + 1$	$-q^{-\phi-\alpha+1}$
T'	e_i , then ribbon	$\alpha = 0$	s	$-q^{-\phi}$

(the sign is that of the corresponding term of the commutator).

Proof. Legality. H: $(b, c) = (a, a + e)$, ribbon first. Need $a \in M$ (true) and $a + e \notin M$, i.e. $\beta = 0$; then $M' = M \setminus \{a\} \cup \{a + e\}$ and the e_i -step needs $a + e \in M'$ (true) and $a + e - 1 \notin M'$, which holds since $a + e - 1 \notin \{a, a + e\}$ (as $e \geq 2$) and $a + e - 1 \notin M$. T: $(b, c) = (a - 1, a)$, ribbon first. Need $a - 1 \in M$ ($\alpha = 1$) and $a + e - 1 \notin M$ (true); then $M' = M \setminus \{a - 1\} \cup \{a + e - 1\}$ and the e_i -step needs $a \in M'$ — true, as $a \notin \{a - 1, a + e - 1\}$ — and $a - 1 \notin M'$, true. H': node first, $c = a + e$: need $a + e \in M$ ($\beta = 1$) and $a + e - 1 \notin M$ (true); then $M'' = M \setminus \{a + e\} \cup \{a + e - 1\}$ and the ribbon step $a \mapsto a + e$ needs $a \in M''$ (true) and $a + e \notin M''$ (true). T': node first, $c = a$: need $a \in M$ (true) and $a - 1 \notin M$ ($\alpha = 0$); then $M'' = M \setminus \{a\} \cup \{a - 1\}$ and the ribbon step $a - 1 \mapsto a + e - 1$ needs $a - 1 \in M''$ (true) and $a + e - 1 \notin M''$ (true).

Heights, by Lemma 3(iii), using $a \in M$, $a + e - 1 \notin M$:

$$\begin{aligned} \text{H} : \#(M \cap [a + 1, a + e - 1]) &= s; & \text{T} : \#(M \cap [a, a + e - 2]) &= 1 + s; \\ \text{T}' : \#(M'' \cap [a, a + e - 2]) &= s \quad (a \notin M''); & \text{H}' : \#(M'' \cap [a + 1, a + e - 1]) &= s + 1 \quad (a + e - 1 \in M''). \end{aligned}$$

Weights, by Lemma 4. Shifting the index of F by one period and using $m(a) = 1$,

$$F_\lambda(a + e) = (m(a - 1) - m(a)) + F_\lambda(a) = \phi + \alpha - 1. \quad (5)$$

H: the e_i -weight is read on the target ν , at content $a + e$, so it is $q^{-F_\nu(a+e)}$. The sum $F(a + e)$ involves m only at positions $\leq a$; $M(\nu)$ differs from M at a (a bead lost, and a does occur, as the $k = 1$ term $-m(a)$) and at $a + e - 1 > a$ (which does not occur). Hence $F_\nu(a + e) = F_\lambda(a + e) + 1 = \phi + \alpha$ by (5). T: the weight is $q^{-F_\nu(a)}$; $F(a)$ involves m only at positions $\leq a - e < a$, and $M(\nu)$ differs from M only at a and $a + e - 1$, both $\geq a$. So $F_\nu(a) = \phi$. T': the weight is $q^{-F_\kappa(a)}$ with $M(\kappa) = M'' = M \setminus \{a\} \cup \{a - 1\}$; again $F(a)$ sees only positions $\leq a - e \leq a - 2$, so $F_\kappa(a) = \phi$. H': the weight is $q^{-F_\kappa(a+e)}$ with $M'' = M \setminus \{a + e\} \cup \{a + e - 1\}$; $F(a + e)$ sees only positions $\leq a$, and both changed positions exceed a , so $F_\kappa(a + e) = F_\lambda(a + e) = \phi + \alpha - 1$. $\square$

Theorem 9 (Closed form, e_i). *Let $e \geq 2$, let λ be a partition and $i \in \mathbb{Z}/e\mathbb{Z}$.*

(a) $([e_i, R_e(t)])_{\nu\lambda} = 0$ unless $\nu \supseteq \lambda$ with ν/λ an $(e-1)$ -ribbon, whose associated bead move $a \mapsto a + e - 1$ satisfies $a \equiv i \pmod{e}$.

(b) For such ν , with α, β, s, ϕ as in (4),

$$([e_i, R_e(t)])_{\nu\lambda} = (\alpha - \beta) q^{-\phi-\alpha} t^s (1 + qt).$$

Proof. (a) is Lemma 6 together with Proposition 7 and the residue observation at the end of the proof of Lemma 6.

(b) By Lemma 8, exactly one of $\{H, H'\}$ is legal (according to β) and exactly one of $\{T, T'\}$ is legal (according to α); so the entry is a sum of exactly two terms. Four cases.

$\alpha = \beta = 0$: legal are H (height s , weight $+q^{-\phi}$) and T' (height s , weight $-q^{-\phi}$). They cancel; and $(\alpha - \beta) = 0$. ✓

$\alpha = \beta = 1$: legal are T (height $s+1$, $+q^{-\phi}$) and H' (height $s+1$, $-q^{-\phi-1+1} = -q^{-\phi}$). They cancel; and $(\alpha - \beta) = 0$. ✓

$\alpha = 1, \beta = 0$: legal are H (height s , $+q^{-\phi-1}$) and T (height $s+1$, $+q^{-\phi}$), giving $q^{-\phi-1}t^s + q^{-\phi}t^{s+1} = q^{-\phi-1}t^s(1+qt)$, and $(\alpha - \beta)q^{-\phi-\alpha} = q^{-\phi-1}$. ✓

$\alpha = 0, \beta = 1$: legal are T' (height s , $-q^{-\phi}$) and H' (height $s+1$, $-q^{-\phi+1}$), giving $-q^{-\phi}t^s(1+qt)$, and $(\alpha - \beta)q^{-\phi-\alpha} = -q^{-\phi}$. ✓ $\square$

6 The f_i case

Everything above dualises. Now $|\nu| = |\lambda| + e + 1$, a path is a pair (b, d) (ribbon $b \mapsto b + e$; added node of content d , i.e. bead move $d-1 \mapsto d$), and the composite is $M \mapsto M \setminus \{b, d-1\} \cup \{b+e, d\}$. Exactly as in Lemma 6: $b = d-1$ and $b+e = d$ are impossible (each would require a bead to be removed twice, resp. adjoined twice), the remaining coincidences are $b+e = d-1$ (type H) and $b = d$ (type T), and both force ν to arise from λ by the single bead move $a \mapsto a + e + 1$ — an $(e+1)$ -ribbon addition — with $a = b$, $d = a + e + 1$ in type H and $a = d-1$, $b = a+1$ in type T . Both types have $d \equiv a+1 \pmod{e}$. When no coincidence occurs the two orders are simultaneously legal, carry the same power of t , and (by the argument of Proposition 7, applied to G instead of F) the same weight, so they cancel.

Set, for $a \in M$ with $a + e + 1 \notin M$,

$$\alpha' := m(a+1), \quad \beta' := m(a+e), \quad s := \#(M \cap [a+2, a+e-1]), \quad \psi := G_\lambda(a+1),$$

so again $0 \leq s \leq e-2$. The analogue of Lemma 8 reads

type	order	legal iff	height	weight
H	ribbon, then f_i	$\beta' = 0$	$s + \alpha'$	$+q^{\psi-\beta'}$
T	ribbon, then f_i	$\alpha' = 1$	$s + \beta'$	$+q^{\psi-1}$
H'	f_i , then ribbon	$\beta' = 1$	$s + \alpha'$	$-q^{\psi-\beta'}$
T'	f_i , then ribbon	$\alpha' = 0$	$s + \beta'$	$-q^\psi$

(For the weights one uses $G_\lambda(a+e+1) = G_\lambda(a+1) - (m(a+e) - m(a+e+1)) = \psi - \beta'$, and the fact that the two positions altered by the other move lie outside the relevant progression except in type T , where the newly created bead at $a+e+1$ contributes -1 .)

Theorem 10 (Closed form, f_i). $([f_i, R_e(t)])_{\nu\lambda} = 0$ unless $\nu \supseteq \lambda$ with ν/λ an $(e+1)$ -ribbon given by the bead move $a \mapsto a + e + 1$ and $i \equiv a + 1 \pmod{e}$; and in that case

$$([f_i, R_e(t)])_{\nu\lambda} = (\alpha' - \beta') q^{\psi-1} t^s (1 + qt).$$

Proof. Identical case analysis. $\alpha' = \beta' = 0$: H (height $s, +q^\psi$) and T' (height $s, -q^\psi$) cancel. $\alpha' = \beta' = 1$: H' (height $s+1, -q^{\psi-1}$) and T (height $s+1, +q^{\psi-1}$) cancel. $\alpha' = 1, \beta' = 0$: T (height $s, +q^{\psi-1}$) and H (height $s+1, +q^\psi$) give $q^{\psi-1} t^s (1 + qt)$. $\alpha' = 0, \beta' = 1$: H' (height $s, -q^{\psi-1}$) and T' (height $s+1, -q^\psi$) give $-q^{\psi-1} t^s (1 + qt)$. $\square$

7 Corollaries

Corollary 11 (Q59, affirmative). For every ν, λ , every i and $x \in \{e, f\}$,

$$\Phi_{\nu\lambda}^{x,i}(t) = 0 \quad \text{or} \quad \varepsilon q^k t^b (1 + qt), \quad \varepsilon = \pm 1, \quad k \in \mathbb{Z}, \quad 0 \leq b \leq e - 2.$$

Moreover $\varepsilon = \alpha - \beta$ (resp. $\alpha' - \beta'$), so that $\Phi \neq 0$ iff exactly one of the two positions $a - 1, a + e$ (resp. $a + 1, a + e$) carries a bead: the commutator is nonzero precisely on the “asymmetric” configurations.

Corollary 12 ($B_{-1}^{[e]}$ is central for $U_q(\widehat{\mathfrak{sl}}_e)$). $[e_i, R_e(-q^{-1})] = [f_i, R_e(-q^{-1})] = 0$ for all i .

Proof. $1 + qt$ vanishes at $t = -q^{-1}$. $\square$

This is a new proof of the level-1 case of the Kashiwara–Miwa–Stern commuting-Heisenberg theorem for the first Heisenberg generator, obtained here as a by-product rather than assumed. It is worth recording that the proof of Theorem 9 nowhere uses it.

Corollary 13 (The commutator with P_e ; (P1),(P2) promoted). $P_e = R_e(-q)$ satisfies

$$([x_i, P_e])_{\nu\lambda} = -\varepsilon(-1)^s q^{k+s+1} (q - q^{-1})$$

in the notation of Corollary 11. In particular every entry of $[e_i, P_e]$ and $[f_i, P_e]$ is either 0 or $\pm q^m (q - q^{-1})$ for a single $m \in \mathbb{Z}$: the quotient by $(q - q^{-1})$ is a unit monomial. Equivalently, the two statements

(P1) the t -support of $\Phi_{\nu\lambda}$ is empty or a pair of consecutive integers, and

(P2) every entry of $[x_i, N_e^{(h)}]$ is 0 or $\pm q^k$,

hold for all $e \geq 2$ and all λ, ν, i .

Proof. $\varepsilon q^k t^s (1 + qt)$ at $t = -q$ equals $\varepsilon q^k (-q)^s (1 - q^2) = -\varepsilon(-1)^s q^{k+s+1} (q - q^{-1})$. (P1) and (P2) are read off the boxed formulas: the support is $\{s, s+1\}$ or empty, and the two coefficients are εq^k and εq^{k+1} . $\square$

Corollary 14 (The commutator with $C_e^{(1)}$). Every entry of $[e_i, C_e^{(1)}]$ and $[f_i, C_e^{(1)}]$ is 0 or $\pm q^m$:

$$([x_i, C_e^{(1)}])_{\nu\lambda} = -\varepsilon(-1)^s q^{k+s+1}.$$

Proof. $C_e^{(1)} = \sum_h (-1)^h [h]_q N_e^{(h)}$, so the entry is $\varepsilon q^k ((-1)^s [s]_q + (-1)^{s+1} q [s+1]_q) = \varepsilon(-1)^s q^k ([s]_q - q[s+1]_q)$, and $[s]_q - q[s+1]_q = \frac{(q^s - q^{-s}) - (q^{s+2} - q^{-s})}{q - q^{-1}} = \frac{q^s(1 - q^2)}{q - q^{-1}} = -q^{s+1}$. $\square$

Corollary 15 (Shape of the defect operator). $[e_i, R_e(t)] = (1 + qt) D_i^-(t)$ and $[f_i, R_e(t)] = (1 + qt) D_i^+(t)$, where $D_i^\mp(t)$ is supported on $(e \mp 1)$ -ribbon additions — in particular on connected skew shapes — with at most one nonzero entry per target, and all its entries unit monomials in q times a power of t .

Example 16. $e = 3$, $\lambda = (2)$, $i = 0$, $x = e_0$. Then $M = \{1, -2, -3, -4, \dots\}$. Taking $a = -3$: $a + e - 1 = -1 \notin M$, so $M(\nu) = M \setminus \{-3\} \cup \{-1\}$, i.e. $\nu = (2, 1, 1)$; and $a \equiv 0 \pmod{3}$. The statistics are $\alpha = m(-4) = 1$, $\beta = m(0) = 0$, $s = \#(M \cap [-2, -2]) = 1$, and $\phi = F_\lambda(-3) = \sum_{k \geq 1} (m(-3k - 4) - m(-3k - 3)) = 0$. So $\Phi_{(2,1,1),(2)}(t) = q^{-1}t(1 + qt) = q^{-1}t + t^2$, which is what the matrix computation returns. By contrast $a = -2$ gives $\nu = (2, 2)$ with $\alpha = m(-3) = 1 = \beta = m(1)$, so $\Phi_{(2,2),(2)} = 0$: the two length-two paths $(2) \rightarrow (2, 2, 1) \rightarrow (2, 2)$ and $(2) \rightarrow (1) \rightarrow (2, 2)$ cancel exactly.

8 Verification

All statements were checked against an independent computation of the matrices of e_i , f_i and $N_e^{(h)}$ built from the definitions (`probes/2026-08-30-Q55-transpose/{ribbons,operators,chevalley}.py`), which in turn rests on two independent enumerations of e -ribbon addition (abacus and direct diagram search) that were checked to agree.

statement	range	result
Lemma 4 (telescope) vs. direct $N^\downarrow, N^\uparrow$	$e \leq 6, \lambda \leq 10$	7070 checks, 0 failures
Theorems 9, 10 vs. matrix $[x, N_e^{(h)}]$	$e \leq 7, \lambda \leq 11$	2737 entries, 0 mismatches
same, extended	$e \leq 9, \lambda \leq 14$	7328 entries, 0 mismatches
support size = 2 on every nonzero entry	$e \leq 9, \lambda \leq 14$	7328/7328
$[x_i, B_{-1}^{[e]}] = 0$	$e \leq 7, \lambda \leq 11$	9074 instances, 0 nonzero
Corollary 13 exact value of $[x_i, P_e]$	$e \leq 7, \lambda \leq 11$	4398 entries, 0 discrepancies
$[x_i, C_e^{(1)}]$ entries unit monomials	$e \leq 7, \lambda \leq 11$	4398 entries, 0 exceptions
conjugation symmetry $b \mapsto e - 2 - b$, $\varepsilon \mapsto \varepsilon$	$e \leq 6, \lambda \leq 9$	858 instances, 0 failures
Lemma 8 and its f_i analogue, <i>per type</i>	$e \leq 7, \lambda \leq 11$	25874 checks, 0 failures

The comparison is two-sided: the test loops over the union of the predicted and the computed supports, so an entry that the closed form fails to predict registers as a mismatch just as a spurious prediction would. Scripts: `probes/2026-08-31-Q59-rigidity/{maya,closed,closed_big,corollaries}.py`.

9 What is and is not claimed

The operator $R_e(t)$ is the spin-graded ribbon Pieri operator of Lascoux–Leclerc–Thibon and is certainly not new; nor is the abacus dictionary of Lemma 3, nor the commutation of $B_{-1}^{[e]}$ with $U_q(\widehat{\mathfrak{sl}}_e)$. What is claimed here is the closed form of Theorems 9 and 10, and the observation that the right language for it is the composition of two bead moves — which reduces the entire question to a four-line case check, with no induction and no case analysis over ribbon shapes.

Two remarks on method, since the target was set up expecting something harder.

1. The session brief proposed a *dimension* argument (“rigidity wants a rank-one space”). That was the wrong instinct here. The rigidity is not the shadow of a one-dimensional space of operators; it is the statement that two bead moves overlap in at most one way, so that at

most two paths survive. The “rank one” is a *cancellation*, and the cancellation is visible only after the geometric “strictly below” in the Chevalley weight is replaced by the arithmetic progression of Lemma 4.

2. The brief also proposed exploiting the functional equation $\Omega R_e(t)\Omega = t^{e-1}R_e(1/t)$ (proved 2026-08-30) to force $\deg_t \Phi \leq b + 1$. That cannot work: combined with $\Omega e_i \Omega = qK_{-i}^{-1}\bar{e}_{-i}$ and the fact that $R_e(t)$ commutes with every K_j (an e -ribbon contains exactly one cell of each residue, so it does not change the $\widehat{\mathfrak{sl}}_e$ -weight), the functional equation relates $\Phi_{\nu'\lambda'}^{e,i}(t)$ to $t^{e-1}\overline{\Phi_{\nu\lambda}^{e,-i}}(1/t)$. It is a symmetry of the family of instances — it sends the instance (ν, λ, i, b) to $(\nu', \lambda', -i, e - 2 - b)$, preserving ε — not a constraint on a single instance, and so it can halve the work but cannot bound a degree. (That this symmetry does hold, exactly as predicted, is a useful independent check on the closed form: 858 instances, 0 failures; see §8.)

There is no gap in the proof of Theorems 9 and 10. The one place where a reader should look hardest is the weight bookkeeping of Lemma 8: the e_i -weight is read on the *target* of the e_i -step (equation (1)), which is ν in types H, T but κ in types H', T'. Getting that backwards exchanges α and β and destroys the two cancellations. For that reason the tables of Lemma 8 and §6 were also verified *row by row* rather than only in aggregate: for each of the eight rows, the legality condition, the height and the weight were compared against the honest Chevalley weight function — not against the telescope of Lemma 4 — on every (λ, a) with $e \leq 7$, $|\lambda| \leq 11$ (25874 checks, 0 failures, `pertype.py`). A compensating pair of errors, which the aggregate test could in principle absorb, would show up there.

Level. Everything above is level 1. The bead-move classification is not obviously level-1-specific — at higher level a bead move happens on one of ℓ runners — and the natural next question is whether Theorems 9 and 10 survive verbatim for Uglov’s level- ℓ Fock spaces, where the Chevalley weight is a sum over runners and the telescope of Lemma 4 acquires ℓ terms.